

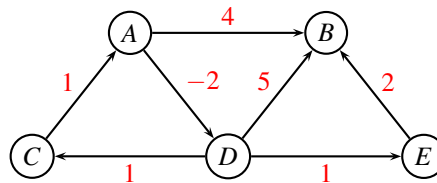
Solution, 17 December, 2020

Course title: Network Optimization

Course no: 42115

Shortest path problem

Consider the following directed graph where distances w_{ij} are marked with red. You are asked to find the shortest path from A to all other nodes.

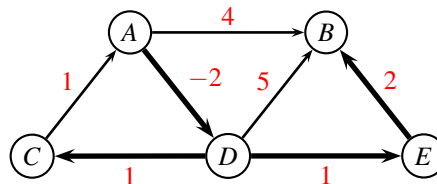


Answer 1 The single negative edge is only part of one cycle, $ADCA$ which has cost 0. So, there are no negative cycles, and the problem is well-defined. We can use Bellman-Ford, or we can use dynamic programming (All edges entering A can be removed, so the remaining problem becomes a DAG). ■

Answer 2 Using Bellman-Ford (considering edges top-down) we get

iteration	A	B	C	D	E
0	0	∞	∞	∞	∞
1	0	4	∞	-2	-1
2	0	1	-1	-2	-1
3	0	1	-1	-2	-1

We terminate after iteration 3, since the values do not change. The shortest-path tree is:



Using dynamic programming, we order the edges in topological ordering A, D, C, E, B , and use the recursion:

$$d_j = \min_{(i,j) \in E} (d_i + w_{ij})$$

In this way we find:

A	D	C	E	B
0	-2	-1	-1	1

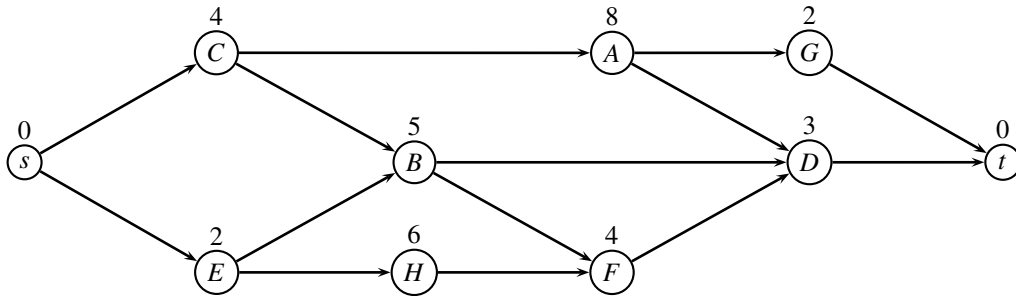
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Project planning

Consider a critical path problem, in which each task i has a processing time p_i , and a list of predecessors P_i . The predecessors P_i need to finish before task i can begin.

task	predecessor(s)	processing time p_i
A	C	8
B	C, E	5
C		4
D	A, B, F	3
E		2
F	B, H	4
G	A	2
H	E	6

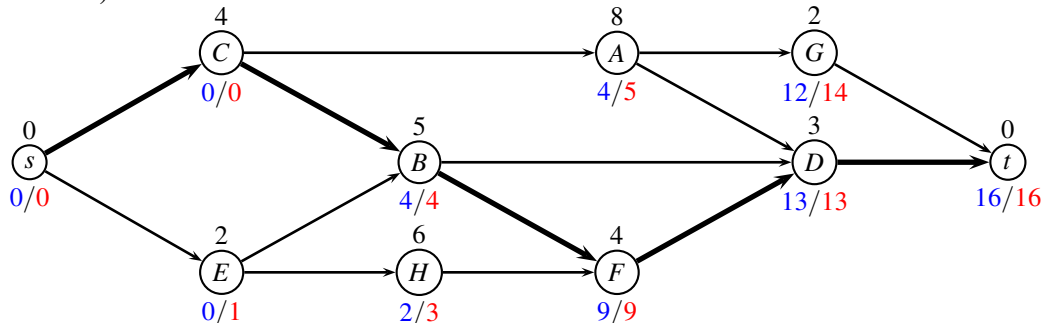
Answer 3 The topological ordering is found by repeatedly selecting a node i with no incoming edges and removing edges leaving i . Moreover, we add activity s before all activities, and t after all activities.



■

Answer 4 Now, we solve the problem

- *Forward procedure:* Starting at time zero $U_s = 0$, calculate the earliest time U_j each activity j can be started $U_j = \max_{(i,j) \in E} (U_i + d_i)$. (marked with blue)
- *Makespan:* $T = U_t = 16$.
- *Backward procedure:* Starting at time equal to the makespan $V_t = T$, calculate the latest time V_j each activity j can be started so that this makespan is realized. $V_i = \min_{(i,j) \in E} (V_j - d_i)$ (marked with red)

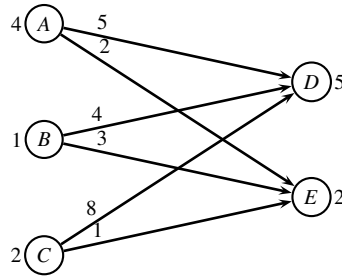


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Answer 5 The critical path is found where $U_i = V_j$. The fat edges are marked on the above figure. ■

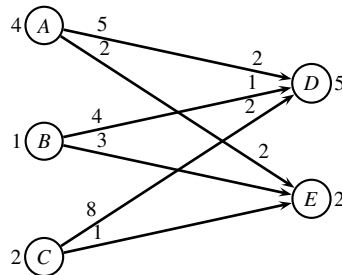
Transportation problem

Three factories A, B, C are transporting items to two customers D, E . The *transportation costs* of sending one item along an edge is indicated on the graph below. The production at factories are respectively $p_A = 4, p_B = 1, p_C = 2$, while the demands at the customers are $d_D = 5, d_E = 2$.



A greedy algorithm considers the factories in the order A, B, C , in each step sending the items to the customers in the cheapest possible way.

Answer 6 Considering the nodes A, B, C we find the greedy solution:

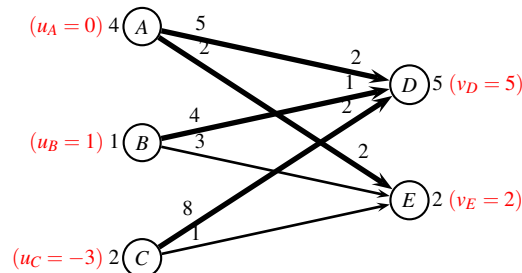


The cost of the greedy solution is

$$z = 5 \cdot 2 + 2 \cdot 2 + 4 \cdot 1 + 8 \cdot 2 = 34$$

■

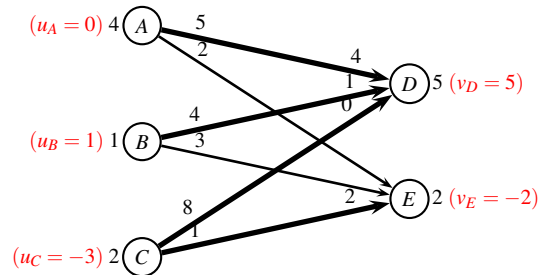
Answer 7 Edges having a flow form a tree (fat edges). We notice that there are $n + m - 1 = 3 + 2 - 1 = 4$ fat edges so the basis has the correct dimension. Calculate the dual variables (red) as the distance from A:



Reduced costs of thin edges is calculated as $\bar{c}_{ij} = c_{ij} + u_i - v_j$, giving

$$\begin{aligned}\bar{c}_{BE} &= 3 + 1 - 2 = 2 \\ \bar{c}_{CE} &= 1 - 3 - 2 = -4\end{aligned}$$

Edge with most negative reduced cost is (C, E) . The edge imposes the cycle C, E, A, D, C . The capacity is $\theta = \min\{2, 2\} = 2$. Sending 2 units along the cycle, we get:



Edge (C, E) enters the tree, while edge (A, E) leaves the tree. We keep edge (C, D) in the tree although the flow is zero.

The dual variables (red) are calculated as distance from A. Reduced costs of thin edges are:

$$\begin{aligned}\bar{c}_{AE} &= 2 + 0 - (-1) = 3 \\ \bar{c}_{BE} &= 3 + 1 - (-1) = 5\end{aligned}$$

No edges have reduced cost, so we have found the optimal solution.

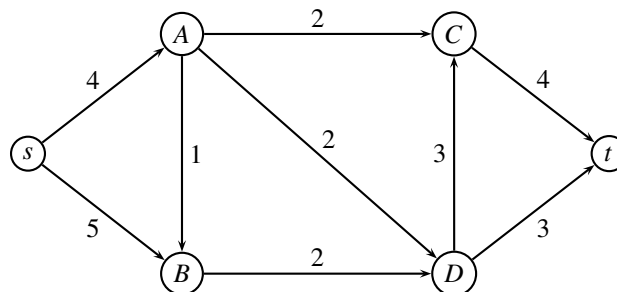
The cost of the optimal solution is

$$z = 5 \cdot 4 + 4 \cdot 1 + 8 \cdot 0 + 1 \cdot 2 = 26$$

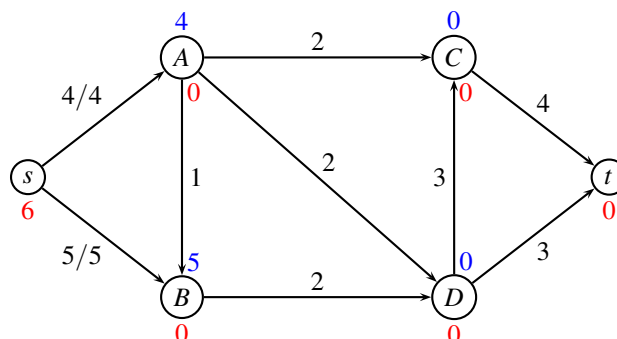
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Maximum flow problem

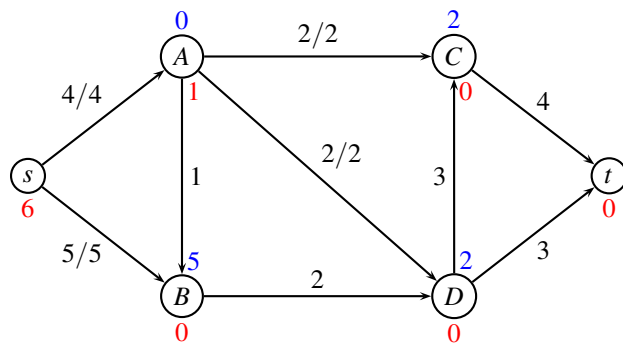
In the following network the edge weights denote the capacity. The problem is to find the maximum flow from s to t .



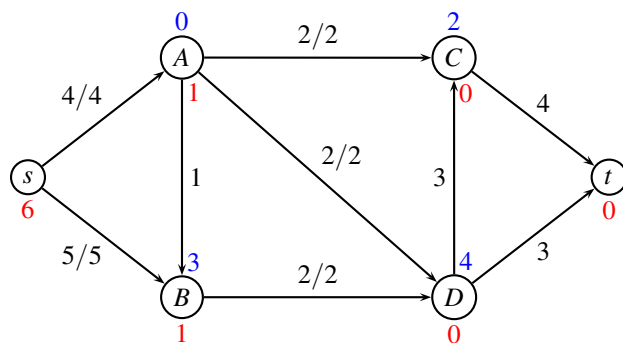
Answer 8 We use preflow-push. Initialize a valid preflow by setting labels $d_s = n = 6$ and remaining $d_i = 0$. (labels d_i are red, surplus $f_x(i)$ are blue)



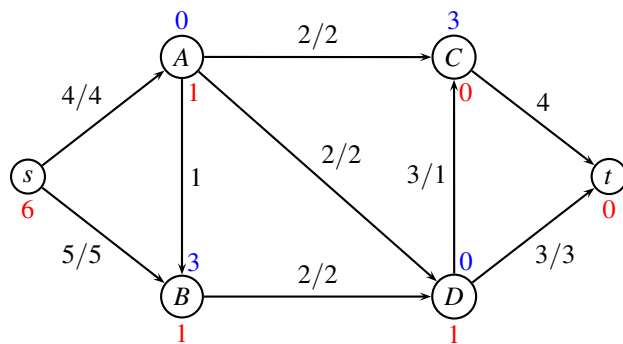
Relabel A to 1, push



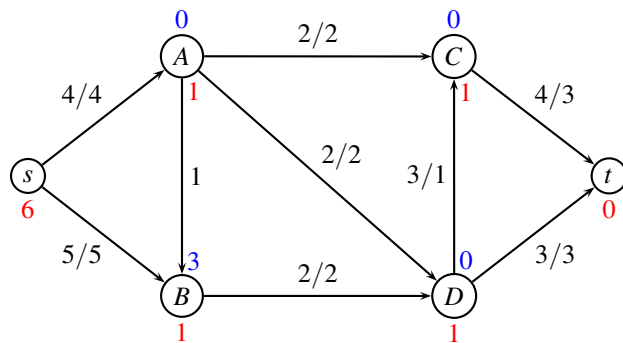
Relabel B to 1, push



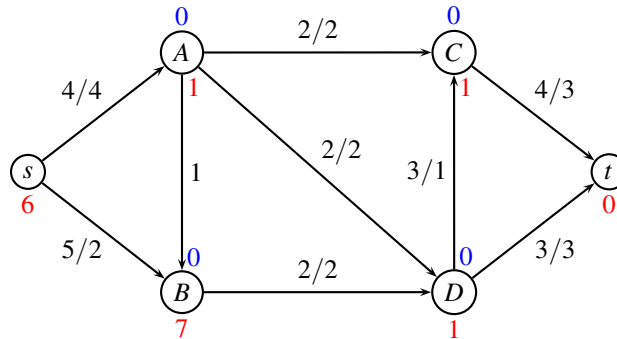
Relabel D to 1, push



Relabel C to 1, push



Relabel B to 7, push

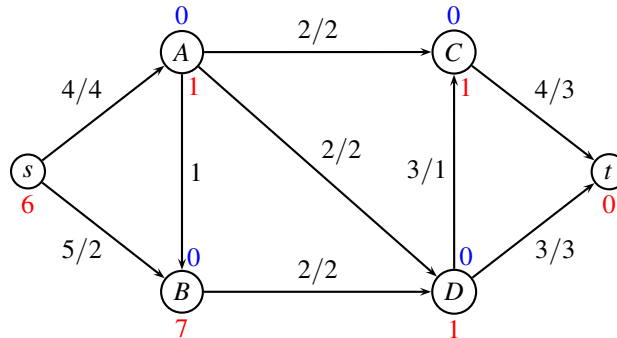


We have a valid flow, and hence the algorithm terminates. The maximum flow is $z = 6$.

If we use augmenting-path we get the following:

- Sending 2 units along path $sACt$
- Sending 2 units along path $sADt$
- Sending 1 unit along path $sBDt$
- Sending 1 unit along path $sBDCt$

This leads to the solution



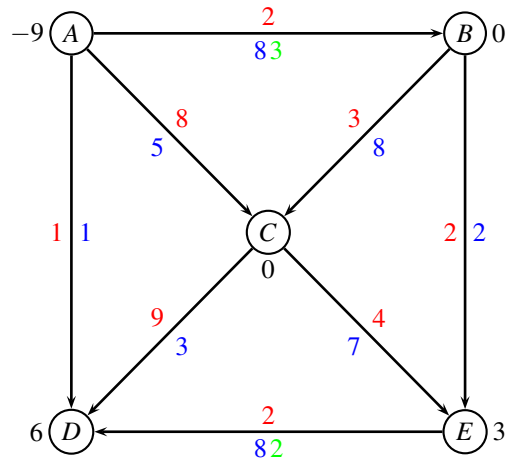
No more augmenting paths exist, so the algorithm stops. ■

Answer 9 If we used the preflow-push method: A minimum cut is found by choosing an unused label, e.g. $k = 2$. We then have $R = \{s, B\}$ and $\bar{R} = \{A, C, D, t\}$. The capacity of the cut is $4 + 2 = 6$.

If we used the augmenting-path method: We mark all nodes red that are reachable from s in the residual graph. These form the cut $R = \{s, B\}$ and $\bar{R} = \{A, C, D, t\}$. The capacity of the cut is $4 + 2 = 6$. ■

Repositioning of containers

Consider the following network with edge costs c_{ij} (red) and edge capacities u_{ij} (blue). Moreover, we have lower bounds ℓ_{ij} (green) on some of the edges. The demand b_i (black) in every node is positive for a sink, and negative for a source.

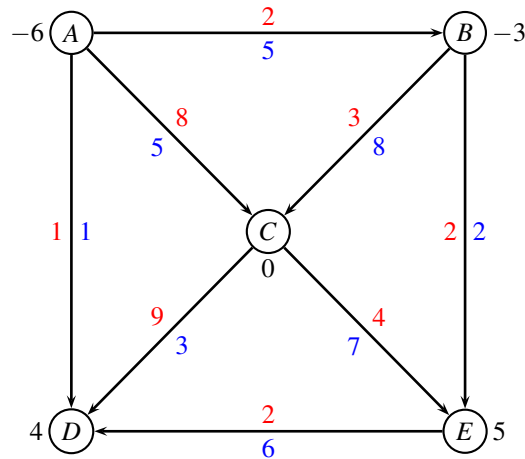


(*)

Answer 10 Assume that edge (i, j) has upper bound u_{ij} and lower bound ℓ_{ij} .

- Replace the upper bound with $u_{ij} := u_{ij} - \ell_{ij}$
- Set the demand at node i to $b_i := b_i + \ell_{ij}$
- Set the demand at node j to $b_j := b_j - \ell_{ij}$

We use the transformation for edges (A, B) and (E, D) getting:

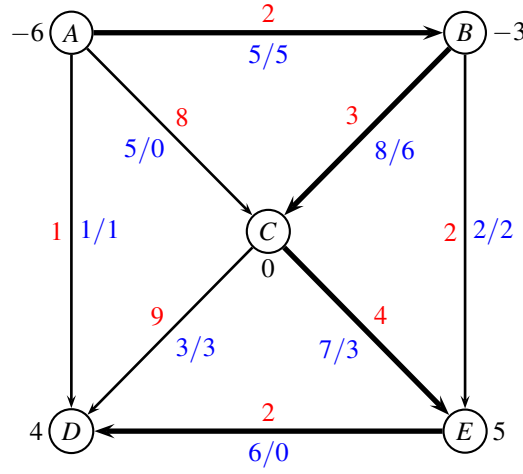


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Now, consider the transformed problem. Having solved a maximum flow problem to find a feasible solution we are told that $T = \{(A, B), (B, C), (C, E), (E, D)\}$ and $U = \{(A, D), (B, E), (C, D)\}$ and $L = \{(A, C)\}$. Edges in T form a tree. The flow on edges in U is at the upper bound, the flow on edges in L is at the lower bound (i.e. zero).

Answer 11 Every partitioning of edges in T, U, L defines a unique flow as shown in the notes. In the

specific case we have



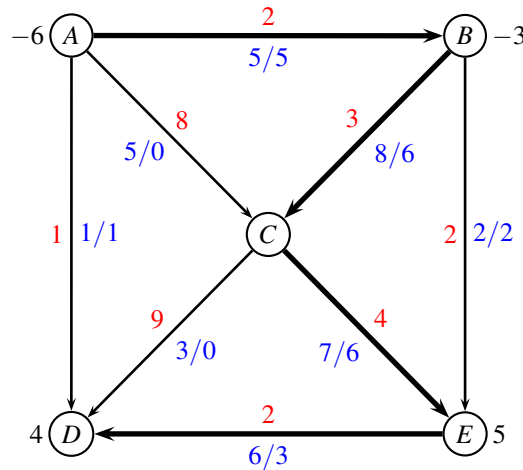
Notice that edges in T are allowed to be at the lower or upper bound as well as any value in-between. The solution value is $z = 2 \cdot 5 + 1 \cdot 1 + 3 \cdot 6 + 2 \cdot 2 + 9 \cdot 3 + 4 \cdot 3 = 72$. ■

Answer 12 We use Network simplex. Choosing A as root node, we find potentials (distances) $y_A = 0$, $y_B = 2$, $y_C = 5$, $y_E = 9$, $y_D = 11$. The reduced costs are found as $\bar{w}_{ij} = w_{ij} + y_i - y_j$, giving

$$\begin{aligned}\bar{w}_{AC} &= 8 + 0 - 5 = 3 && L \text{ edge} \\ \bar{w}_{AD} &= 1 + 0 - 11 = -10 && U \text{ edge} \\ \bar{w}_{BE} &= 2 + 2 - 9 = -5 && U \text{ edge} \\ \bar{w}_{CD} &= 9 + 5 - 11 = 3 && U \text{ edge}\end{aligned}$$

Since edge (C, D) is a U edge and it has positive reduced cost, we can improve the solution. The edge (C, D) imposes the cycle D, C, E, D (we are going in the backward direction of edge (C, D) since it is a U edge).

The capacity we can send along the cycle is $\theta = \min\{3, 4, 6\} = 3$. Sending three items along cycle D, C, E, D means that edge (C, D) becomes an L edge while the tree T is unchanged.

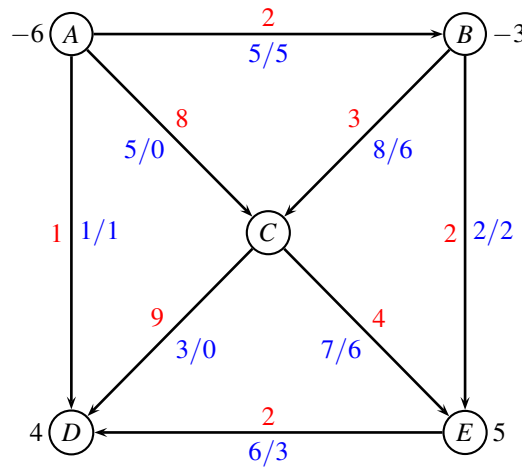


Since T is unchanged we find the same potentials (distances) $y_A = 0$, $y_B = 2$, $y_C = 5$, $y_E = 9$, $y_D = 11$. The reduced costs are also the same (but edge (C, D) is now an L edge)

$$\begin{aligned}
\bar{w}_{AC} &= 8 + 0 - 5 = 3 && L \text{ edge} \\
\bar{w}_{AD} &= 1 + 0 - 11 = -10 && U \text{ edge} \\
\bar{w}_{BE} &= 2 + 2 - 9 = -5 && U \text{ edge} \\
\bar{w}_{CD} &= 9 + 5 - 11 = 3 && L \text{ edge}
\end{aligned}$$

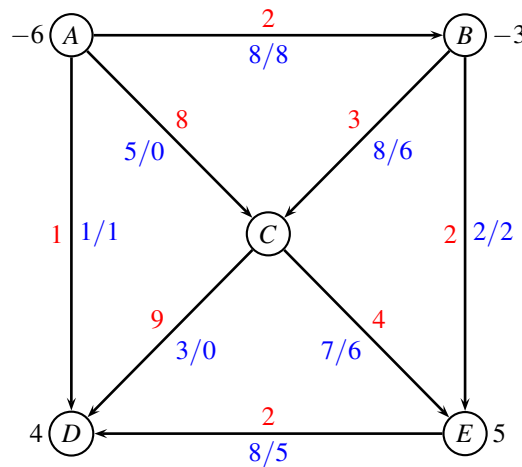
Since all U edges have negative reduced cost, and all L edges have positive reduced cost, we have found the optimal solution. The solution value is $z = 2 \cdot 5 + 1 \cdot 1 + 3 \cdot 6 + 2 \cdot 2 + 4 \cdot 6 + 2 \cdot 3 = 63$.

If we use the augmenting cycle method: Starting from the same initial solution, we find the augmenting cycle C, E, D, C with cost $4 + 2 - 9 = -3$. The capacity we can send along the cycle is $\theta = \min\{4, 6, 3\} = 3$. Sending 3 units along the cycle gives:



No more cycles with negative cost can be found, so the algorithm terminates. ■

Answer 13 We add the lower bounds to all edges $x_{ij} := x_{ij} + \ell_{ij}$ getting the solution



The solution value is $z = 2 \cdot 8 + 1 \cdot 1 + 3 \cdot 6 + 2 \cdot 2 + 4 \cdot 6 + 2 \cdot 5 = 73$.

Checking solution in CPLEX:

```
minimize
2 xAB + 8 xAC + 1 xAD + 3 xBC + 2 xBE + 9 xCD + 4 xCE + 2 xED

subject to
- xAB - xAC - xAD = -9
xAB - xBC - xBE = 0
xAC + xBC - xCD - xCE = 0
xAD + xCD + xED = 6
xBE + xCE - xED = 3

xAB <= 8
xAC <= 5
xAD <= 1
xBC <= 8
xBE <= 2
xCD <= 3
xCE <= 7
xED <= 8

xAB >= 3
xED >= 2

end

Dual simplex - Optimal: Objective = 7.3000000000e+01
Solution time = 0.01 sec. Iterations = 2 (2)
Deterministic time = 0.02 ticks (1.47 ticks/sec)

CPLEX> dis sol var -
Variable Name          Solution Value
xAB                    8.000000
xAD                    1.000000
xBC                    6.000000
xBE                    2.000000
xCE                    6.000000
xED                    5.000000
All other variables in the range 1-8 are 0.
```

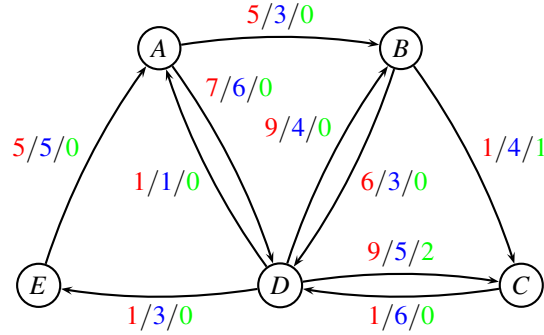
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Flowing containers in a liner shipping network

The below instance, BALTICLINE, shows a small liner shipping network. Nodes represent ports, and edges represent legs sailed by vessels. The capacities indicate the remaining capacity of the vessels after some mandatory orders have been fulfilled. Therefore, the capacity may vary for each leg. **Note:** The network is the same as in *Project Assignment 2020*, but the (source,terminal) pairs for each commodity is changed.

In mathematical terms, we are given a weighted oriented graph $G = (V, E, c, u, \ell)$ where each edge $(i, j) \in E$ has an associated cost (per container) $c_{ij} \geq 0$, a capacity $u_{ij} \geq 0$, and a lower bound $\ell_{ij} \geq 0$.

BALTICLINE (Route net of a Baltic liner shipping company)



For edge (i, j) the first number (red) is the cost c_{ij} , the second number (blue) is the capacity u_{ij} , and the third number (green) is the lower bound ℓ_{ij} .

We have the demands

k	s_k	t_k	d_k
1	D	C	2
2	D	B	1
3	A	E	3

Commodity $k = 1$ is shipping two containers from D to C , commodity $k = 2$ is shipping one container from D to B , and commodity $k = 3$ is shipping three containers from A to E . The commodities should be transported *unsplittable*, i.e. all d_k units should follow the same path.

Answer 14 We have the routes r outlined in the below table. The cost of sending d_k units along a route r is calculated as $d_k \sum_{(i,j) \in r} c_{ij}$.

route r	commodity k	nodes in route	d_k	$\sum c_{ij}$	cost
1	1	DBC	2	10	20
2	1	DC	2	9	18
3	1	$DEABC$	2	12	24
4	2	DAB	1	6	6
5	2	DB	1	9	9
6	2	$DEAB$	1	11	11
7	3	$ABCDE$	3	8	24
8	3	$ABDE$	3	12	36
9	3	ADE	3	8	24

■

Answer 15

We introduce the binary variable x_i for route number i . Using this notation the path formulation for BALTICLINE becomes:

```

minimize
+20x1+18x2+24x3+6x4+9x5+11x6+24x7+36x8+24x9
subject to
UAB:      +2x3+1x4      +1x6+3x7+3x8      <= 3
UAD:      +3x9 <= 6
UBC: +2x1      +2x3      +3x7      <= 4
UBD:      +3x8 <= 3
UCD:      +3x7 <= 6
UDA:      +1x4 <= 1
UDB: +2x1      +1x5      <= 4
UDC:      +2x2 <= 5
UDE:      +2x3      +1x6+3x7+3x8+3x9 <= 3
UEA:      +2x3      +1x6 <= 5

LBC: +2x1      +2x3      +3x7      >= 1
LDC:      +2x2 >= 2

K1: +x1+x2+x3 = 1
K2: +x4+x5+x6 = 1
K3: +x7+x8+x9 = 1

binary
x1 x2 x3 x4 x5 x6 x7 x8 x9
end

CPLEX> opt
MIP - Integer optimal solution: Objective = 5.1000000000e+01

CPLEX> dis sol var -
Incumbent solution
Variable Name      Solution Value
x2                  1.000000
x5                  1.000000
x7                  1.000000
All other variables in the range 1-9 are 0.

```

■

Now, assume that we want to solve the LP-relaxed version of the path formulation with column generation. After a few iterations of the algorithm (using dummy routes as initial columns) we have the following dual solution:

Dual variables for the upper bound constraints: $y_{AB} = -2$, $y_{DA} = -3$. Dual variables for the lower bound constraints: $y'_{BC} = 7$, $y'_{DC} = 2$. Dual variables for the commodity constraints: $\bar{y}_1 = 8$, $\bar{y}_3 = 2$. All other dual variables are zero.

Answer 16 The reduced cost of a route is calculated as

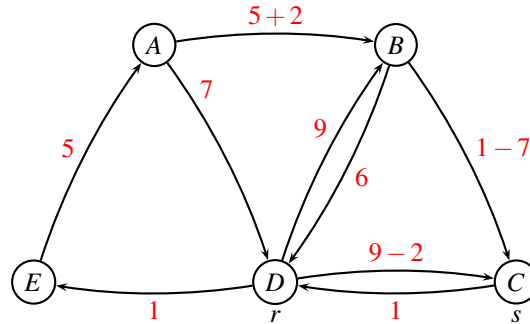
$$\bar{c}_r^k = \sum_{(i,j) \in r} d_k c_{ij} - \sum_{(i,j) \in r} d_k y_{ij} - \sum_{(i,j) \in r} d_k y'_{ij} - \bar{y}_k$$

Which can be written as

$$d_k \sum_{(i,j) \in r} (c_{ij} - y_{ij} - y'_{ij}) - \bar{y}_k$$

We therefore first remove edges with capacity less than d_k , in our case edge (D,A) . Next, we set the

edge costs to $c'_{ij} = c_{ij} - y_{ij} - y'_{ij}$. In this way we get the graph:



Bonus info: The shortest path from D to C has cost $z = 3$. The reduced cost is then

$$\bar{c}_r^k = d_k \cdot z - \bar{y}_k = 2 \cdot 3 - 8 = -2$$

■

Organizing a conference

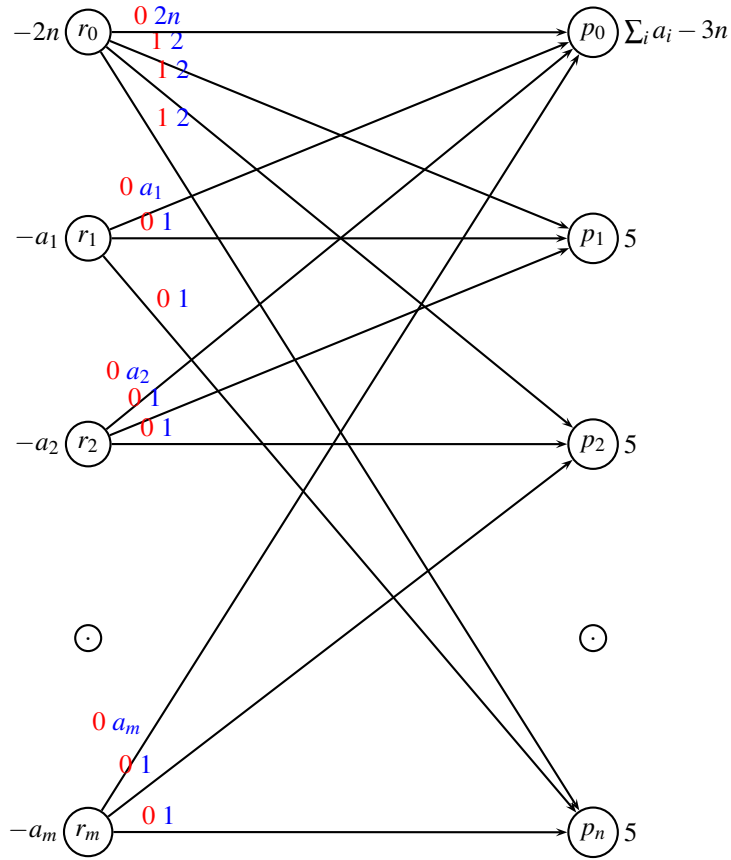
Suppose you are organizing a conference where researchers present articles they have written. Researchers who want to present an article send a paper to the conference organizers. The conference organizers have access to a committee of reviewers $R = \{r_1, \dots, r_m\}$. Each reviewer r_i is willing to read up to a_i articles. There are $P = \{p_1, \dots, p_n\}$ paper submissions, and each paper must be reviewed by 3 to 5 distinct reviewers (more reviewers is better). Moreover, each paper submission has a particular topic and each reviewer has a specialization for a set of topics, so papers on a given topic only get reviewed by those reviewers who are experts on that topic. The conference organizers need to decide which reviewers will review each article.

Answer 17 We formulate the problem as a *minimum cost flow problem* defined on a bi-partite graph consisting of nodes R and P . We add an edge with capacity 1 and cost 0 from reviewer r_i to paper p_j if the reviewer is qualified to read the paper. Each paper p_j has a demand of 5. Each reviewer r_i has a demand of $-a_i$.

A dummy reviewer r_0 is used to fill reviewers up to the 5 needed. We add an edge from dummy reviewer r_0 to each paper p_j having capacity 2 and cost 1. Dummy reviewer r_0 has demand $-2n$.

A dummy paper p_0 is used to take the surplus of reviewers. We add an edge from each reviewer r_i to dummy paper p_0 having capacity a_i and cost 0. Dummy paper p_0 has demand $(\sum_i a_i + 2n) - 5n = \sum_i a_i - 3n$.

Finally, we add an edge from the dummy reviewer r_0 to the dummy paper p_0 with capacity $2n$ and cost 0. This edge takes a possible surplus of dummy reviewers.



After having solved the minimum cost flow problem, the *assignment of reviewers to papers* is given by the flow on edges from R to P . If no solution can be found, it is not possible to satisfy the constraints. In this case, the capacity on edges from the dummy reviewer r_0 to each paper p_j can be increased, and the model re-solved. This will mean that fewer reviewers may be assigned to each paper. ■

end of assignment